

CM2025 Cyber Security

Course Description

Cybersecurity is a key area of study in theoretical and applied Computer Science. This module imparts key cybersecurity principles such as encryption and authentication, essential for evaluating and enhancing the security of computing applications. Through real-world security scenarios, you will apply and assess core computer science concepts, honing their problem-solving and critical evaluation skills. The module fosters independence as you navigate complex cybersecurity issues, learning how to produce organised work. Additionally, it provides a platform for substantial work from problem inception to implementation, equipping you with the practical skills crucial for addressing contemporary cybersecurity challenges.

This cybersecurity module provides a structured exploration into the domain of digital security. Drawing upon engineering, computational and mathematical foundations, the curriculum enables you to identify and categorise cyber threats, address network security challenges and implement cryptographic methods for data protection. The module also introduces blockchain technology, highlighting its relevance in enhancing transactional security. A notable aspect of the module is its emphasis on the broader societal and ethical dimensions of cybersecurity, prompting you to assess practices in light of environmental and equality, diversity and inclusion (EDI) considerations.

The module consists of the following topics:

1. Identification and classification of cyber threats
2. Cybersecurity frameworks, standards, legal, ethical and professional considerations
3. Network and OS security
4. Application security principles: testing and debugging
5. Cryptography in practice
6. Blockchain for cybersecurity
7. Incident response and forensics
8. Emerging trends
9. EDI principles, sustainable digital ecosystems and employability in cybersecurity
10. Real-world cybersecurity case studies: design, test and communicate security solutions.

Course Goals and Objectives

Upon successful completion of this course, you will be able to:

1. Apply knowledge of engineering and computational principles to identify and classify various cyber threats based on severity and impact.
2. Analyse network security challenges using principles of mathematics and engineering to select and apply appropriate security measures such as configuring firewalls and setting up Intrusion Detection Systems (IDSs).
3. Understand mathematical and computational techniques to implement basic cryptographic algorithms ensuring data integrity and confidentiality.
4. Design blockchain solutions for enhancing cybersecurity in areas like secure transactions, considering societal, user and business needs as well as ethical and inclusivity principles.
5. Evaluate the environmental, societal and ethical impact of current cybersecurity practices, and propose improvements considering the principles of EDI.
6. Design a cybersecurity solution for a real-world problem, test its effectiveness, ensure it aligns with EDI principles and communicate the findings effectively to both technical and non-technical audiences.

Textbook and Readings

This module makes use of a wide range of reading material which you will encounter as you work through the course. The material is available via the online library, and it includes journal and magazine articles as well as some textbook references. There will also be discussion prompts asking you to do some independent research using online sources.

Course Outline

The course consists of ten topics divided into 20 weeks that focus on key concepts.

Topic 1: Identification and classification of cyber threats	Learning outcomes: • Define and explain cyber threats. • Identify different types of cyber threats. • Classify cyber threats based on their nature, source and impact.
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Topic 2: Cybersecurity frameworks, standards, legal, ethical and professional considerations

Learning outcomes:

- Explain the role and scope of cybersecurity standards and internal policies in supporting organisational security objectives.
- Demonstrate understanding of the legal, ethical and professional responsibilities involved in cybersecurity practice.
- Apply cybersecurity frameworks and ethical reasoning to assess and respond to complex security challenges.

Topic 3: Network and OS security	Learning outcomes: • Explain how network and operating system layers interact and describe their security features, vulnerabilities and role in supporting secure digital infrastructures. • Evaluate the effectiveness of system-level security measures, including intrusion detection, prevention, access control and audit tools, through case studies and applied contexts. • Apply practical techniques in network monitoring and OS hardening to analyse and respond to security threats.
Topic 4: Application security principles: testing and debugging	Learning outcomes: • Identify common application security vulnerabilities and explain why they matter in software development. • Apply secure coding principles to mitigate common security threats. • Demonstrate testing and secure debugging practices.

Topic 5: Cryptography in practice	Learning outcomes: • Define cryptography and the importance of its management. • Compare different types of encryption and how they are used. • Apply common cryptographic techniques. • Describe different types of cybersecurity attacks, including brute force, dictionary, man-in-the-middle and side-channel attacks.
Topic 6: Blockchain for cybersecurity	Learning outcomes: • Explain how blockchain secures peer-to-peer transactions and supports applications such as smart contracts. • Analyse the role of blockchain in identity management, including the benefits and challenges of self-sovereign identity. • Evaluate the ethical, regulatory and social implications of blockchain adoption. • Explore emerging trends in blockchain cybersecurity, including AI, IoT, quantum

	cryptography and scalable consensus protocols.
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Topic 7 Incident response and forensics	Learning outcomes: • Describe and apply the phases of the NIST incident response lifecycle. • Explain core digital forensic practices and their role in preserving evidence. • Use frameworks and tools (e.g. cyber kill chain, MITRE ATT&CK, forensic tools) to investigate cyber incidents.
Topic 8: Emerging trends	Learning outcomes: • Analyse how cybersecurity uses AI and machine learning for both offensive and defensive purposes, particularly in cryptographic and blockchain contexts. • Evaluate the impact of quantum computing on classical cryptography and explain how post-quantum algorithms (e.g. Kyber, Dilithium) address emerging threats. • Assess IoT security challenges and apply appropriate cryptographic approaches,

	including lightweight algorithms, to secure constrained devices.
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Topic 9: EDI principles, sustainable digital ecosystems and employability in cybersecurity	Learning outcomes: • Explain and apply equality, diversity and inclusion (EDI) principles in cybersecurity. • Evaluate approaches to building sustainable digital ecosystems, balancing security with environmental, social and ethical considerations. • Develop strategies for employability in cybersecurity, including identifying valued certifications, planning career development and integrating ethical and inclusive practices into your professional pathway.
Topic 10: Real-world cyber security case studies: design, test and communicate security solutions	Learning outcomes: • Explain the principles of cybersecurity in critical infrastructure systems, with a focus on industrial control systems (ICS) and energy networks. • Analyse cyber-attack detection and mitigation strategies in operational technology environments, including smart monitoring and automated response. • Evaluate the challenges of securing renewable energy and smart grid infrastructures, and

	propose solutions that balance resilience, sustainability and reliability.
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Learning Activities of This Course

The course is comprised of the following elements:

- **Lecture videos.** In each topic, you will find a sequence of videos.
- **Readings.** Each topic may include several suggested readings. These are a core part of your learning, and, together with the videos, will cover all the concepts you need for this module.
- **Practice Quizzes.** Each topic will include practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.
- **Discussion Prompts.** Each topic includes discussion prompts. You will see the discussion prompt alongside other items in the lesson. Each prompt provides a space for you to respond. After responding, you can see and comment on your peers' responses. All prompts and responses are also accessible from the general discussion forum and the module discussion forum.

How to Pass This Course

The module will contain a range of summative and formative assessments. Summative assessments are assessments which contribute directly towards your final grade. Formative assessments do not count directly towards your final grade. Instead, they provide you with opportunities for low stakes practice and will often provide some sort of feedback about your progress. For example, a practice quiz might provide you with feedback about why a particular answer was wrong.

The module has two assessments: a midterm assessment and a final exam. The midterm assessment comprises some written research work and some cryptography work. The exam comprises a quiz section and a written section. The midterm and exam each carry 50% of the grade.

This is a detailed breakdown of all the marks:

Activity	Required? (Summative)	Deadline week	Estimated time per course	% of final grade
Written, staff graded coursework	Yes	Approximately week 13	20 hours	50%
Written examination	Yes	Approximately week 22	3-4 hours	50%